

GPT: the Decoder-Only Transformer

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Activation functions ([Note 1027](#)); Masked self-attention ([Note 1081](#)); Transformer ([Note 1083](#)); Transformer decoder ([Note 1083](#)).
- **Leads to:** MLP blocks as fact storage ([Note 1089](#)).

1. Overview

Key point: GPT (G-855) is a transformer with the encoder removed. What is left is a stack of identical blocks, each with two parts: masked self-attention, then an MLP. Each part adds its result to the token's vector. One pass reads a text and gives a guess for the next token at **every** position. Training improves all these guesses at once. Generation keeps only the last guess, appends it, and runs the pass again. We load the released GPT-2 small (124,439,808 parameters) and run it in plain NumPy to check every claim.

The [transformer decoder Note](#) built the decoder of the translation transformer. It has three sub-layers: masked self-attention, cross-attention to the encoder, and a feed-forward network. GPT keeps the decoder and drops the encoder. With no encoder there is nothing to cross-attend to, so the cross-attention goes too. Radford et al. (2018, §4.1) describe their model, a **decoder-only transformer** (G-565), as a “12-layer decoder-only transformer with masked self-attention heads”.

Figure 1 follows one real pass of GPT-2 small on “Steve Jobs was the founder of”. Watch three things:

1. Each token takes its column from the embedding matrix.
2. In attention, arrows only point **right**: a token reads only the tokens before it.
3. At the top, a guess appears above **every** token, not just the last one. The last guess, “Apple”, is the one used for generation.

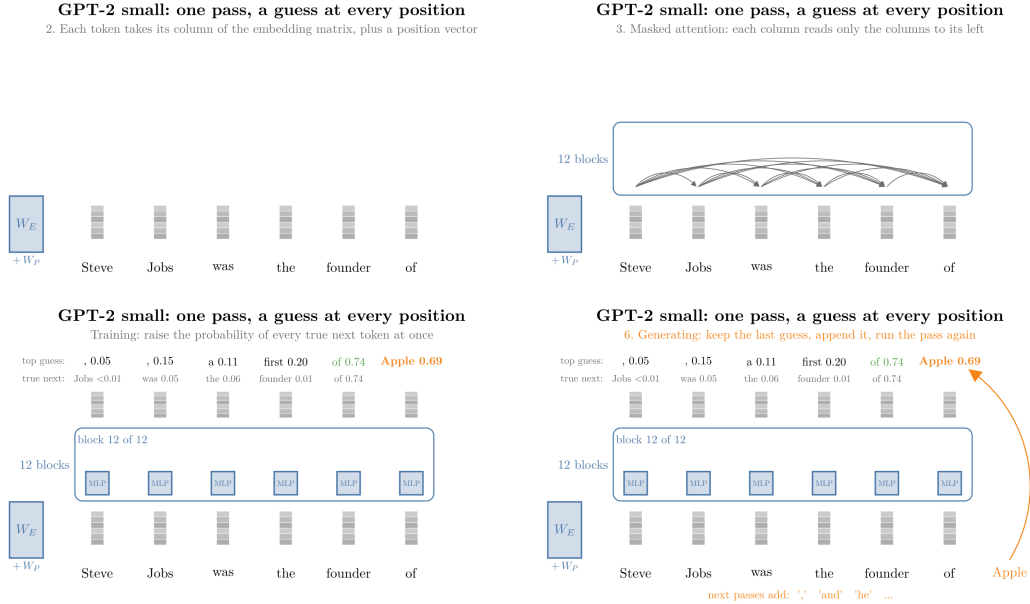


Figure 1: One pass of GPT-2 small (real weights). Blue: learned weights. Grey: the data flowing through. Top row: the model’s top guess for the next token at each position, with its probability. Second row: the true next token and the probability the model gave it (green = guessed right)

The order of this Note follows the data through the model: tokens (section 4), vectors (section 5), the blocks (section 6), the guesses at every position (section 7), how much text fits (section 8), and finally where the parameters sit (section 9).

2. Prerequisites

- [Transformer decoder Note](#): the decoder block, the output layer (linear and softmax), training on every position in one pass.
- [Masked self-attention Note](#): the causal mask; why one pass equals many steps.
- [Transformer inference Note](#): generation one token at a time.
- [Transformer encoder Note](#): residual connections and the feed-forward network.
- [Layer normalisation Note](#): LayerNorm, and the prenorm and postnorm placements.
- [Positional encoding Note](#): why the model needs positions.
- [Multi-head attention Note](#): heads, W_Q , W_K , W_V , W_O , and the $4d^2 + 4d$ count.
- [History of LLMs Note](#): where GPT, GPT-2 and GPT-3 come from.

3. From the transformer decoder to GPT

Key point: GPT keeps the decoder’s masked self-attention and feed-forward network and removes the cross-attention, because there is no encoder to attend to. Each GPT block has two sub-layers instead of three.

The translation transformer needs two inputs: the source sentence (read by the encoder) and

the target written so far (read by the decoder). A language model has only one input: the text so far. It predicts the next token of that same text. So one stack is enough. Radford et al. (2018, §3.1) train it to maximise the likelihood of each token given the k tokens before it:

$$L_1(U) = \sum_i \log P(u_i \mid u_{i-k}, \dots, u_{i-1})$$

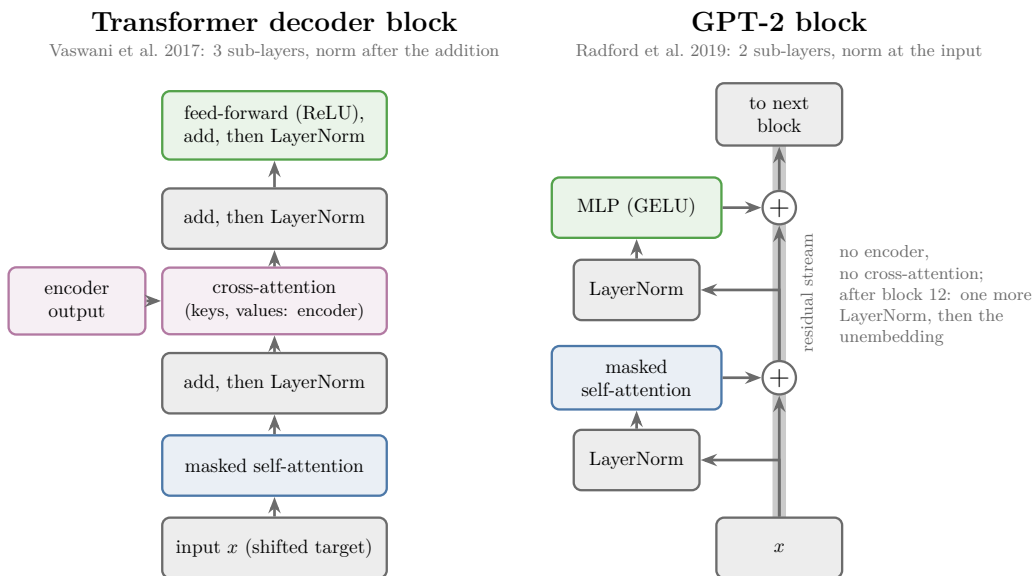


Figure 2: Left: the decoder block of the translation transformer (3 sub-layers, LayerNorm after each addition). Right: a GPT-2 block (2 sub-layers, LayerNorm at the input of each, the residual stream passes straight through)

	Transformer decoder (Vaswani et al. 2017)	GPT-2 block (Radford et al. 2019)
Sub-layers	masked self-attention, cross-attention, feed-forward	masked self-attention, MLP (feed-forward)
Encoder	yes	none
Where LayerNorm sits	after each addition (postnorm)	at the input of each sub-layer (prenorm), plus one after the last block
Activation in the feed-forward network	ReLU	GELU
Positions	fixed sine-cosine vectors	a learned table
Output layer	its own matrix (tied to the embeddings in the paper, §3.4)	the token embedding matrix itself (tied)

Figure 2 shows the two blocks side by side. The rest of this Note goes through the right-hand column, row by row, on the real GPT-2 small. The two names “MLP” and “feed-forward network” mean the same block; GPT-2’s weights call it `mlp`.

4. Tokens

Key point: GPT-2 reads text as **tokens** (G-1981), pieces of words from a fixed list of 50,257. Common words are one token; rare words are split into pieces.

A tokenizer cuts the text into pieces from a fixed vocabulary. GPT-2’s vocabulary has 50,257 tokens (Radford et al. 2019, §2.3). It is built with **byte pair encoding** (G-335), which starts from single bytes and merges the most frequent pairs (Radford et al. 2019, §2.2). The Notebook runs GPT-2’s own tokenizer:

Text	Tokens
Steve Jobs was the founder of	Steve, Jobs, was, the, founder, of (6 tokens)
Unbelievably, tokenization splits rare words.	Un, bel, iev, ably, ,, token, ization, splits, rare, words, . (11 tokens)

The space before a word belongs to the token: “ Jobs” with a space and “Jobs” without one are different tokens (Figure 3). In the other figures we draw the word without its space.

“Steve Jobs was the founder of”: 6 tokens

Steve □Jobs □was □the □founder □of

“Unbelievably, tokenization splits rare words.”: 11 tokens

Un bel iev ably , □token ization □splits □rare □words .

coloured: one rare word split into several tokens; □ marks the space that belongs to the token

Figure 3: GPT-2’s tokenizer on the two example texts. Each box is one token; the open-box sign marks the space that belongs to it. “Unbelievably” becomes 4 tokens and “tokenization” 2

5. From tokens to vectors: token and position embeddings

Key point: Each token’s vector is its row of the **token embedding matrix** (G-1980) W_E ($50,257 \times 768$) plus the row of the **position embedding matrix** (G-1526) W_P ($1,024 \times 768$) for its place in the text. Both matrices are learned.

The first step is a lookup, as in the [what is self-attention Note](#). Token number 19206, “Steve”, takes row 19206 of W_E : 768 numbers. Then the vector for position 0 is added from W_P . For the token t_i at position i :

$$x_i = W_E[t_i] + W_P[i]$$

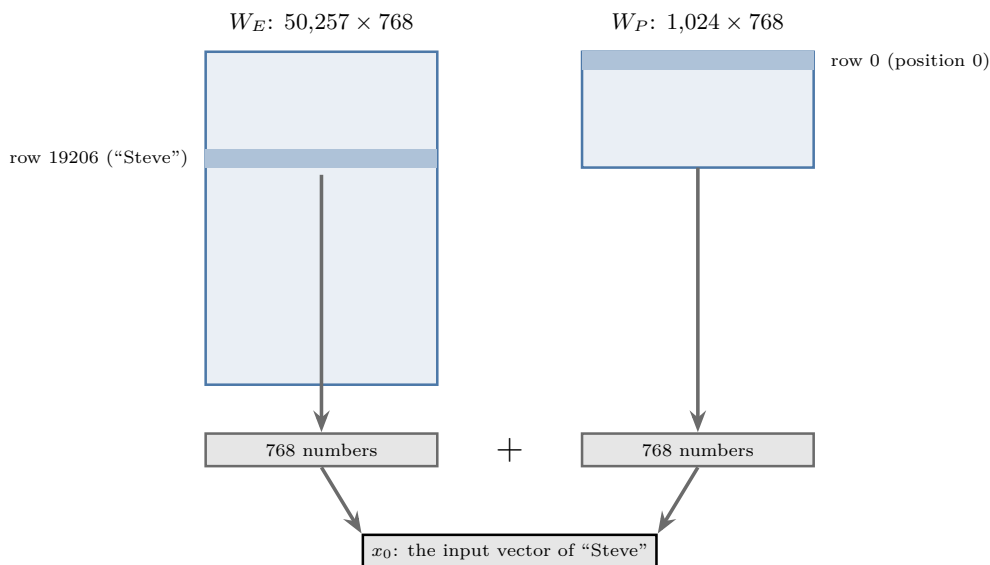


Figure 4: The first input vector of the pass: row 19206 of W_E (the token “Steve”) plus row 0 of W_P (position 0). Blue: learned weights; grey: data

Figure 4 shows the lookup and the addition for the first token.

GPT uses **learned** positions: “We used learned position embeddings instead of the sinusoidal version proposed in the original work” (Radford et al. 2018, §4.1). W_P is just another weight matrix, trained with the rest. The [positional encoding Note](#) explains why positions are needed at all: attention by itself ignores word order.

Extra: Sanderson (2024, Ch 5) draws W_E with one *column* per token; the released GPT-2 file stores one *row* per token. It is the same matrix, transposed. We follow the file.

6. Inside a GPT-2 block

Key point: A block reads the token vector through a LayerNorm, computes a change, and **adds** the change back: $x \leftarrow x + \text{Attn}(\text{LN}_1(x))$, then $x \leftarrow x + \text{MLP}(\text{LN}_2(x))$. The vector that flows from block to block, never replaced, only added to, is called the **residual stream** (G-1683).

6.1 Prenorm: LayerNorm at the input

Key point: GPT-2 moved each LayerNorm from after the addition to the input of the sub-layer, and added one more LayerNorm after the last block.

The paper states the change in one sentence: “Layer normalization (Ba et al., 2016) was moved to the input of each sub-block, similar to a pre-activation residual network (He et al., 2016) and an additional layer normalization was added after the final self-attention block” (Radford et al. 2019, §2.3). The released weights match: every block has `ln_1` (before attention) and `ln_2` (before the MLP), and there is one `ln_f` after block 12. The [layer normalisation Note](#) calls this placement **prenorm**.

1. **In words:** normalise a copy of the vector, let the sub-layer compute a change from the copy, and add the change to the original.
2. **Formula:** for each block,

$$x \leftarrow x + \text{Attn}(\text{LN}_1(x)), \quad x \leftarrow x + \text{MLP}(\text{LN}_2(x))$$

After block 12: $h = \text{LN}_f(x)$.

3. **Example:** the Notebook normalises the stream vector of ” of” before block 6 and gets mean 0.000 and standard deviation 1.000, whatever the stream’s own size.

6.2 The residual stream grows, and the final LayerNorm brings it back

Key point: In GPT-2 small the stream vector of ” of” grows from length 4.9 to 400 over the 12 blocks, because each block adds and nothing rescales. The final LayerNorm brings it back to a fixed scale; without it, the prediction is lost.

In the prenorm formula only the copies are normalised; the stream x itself is never rescaled. Each block adds two vectors to it, so nothing stops its length from growing. Figure 5 measures this for the last token, ” of”:

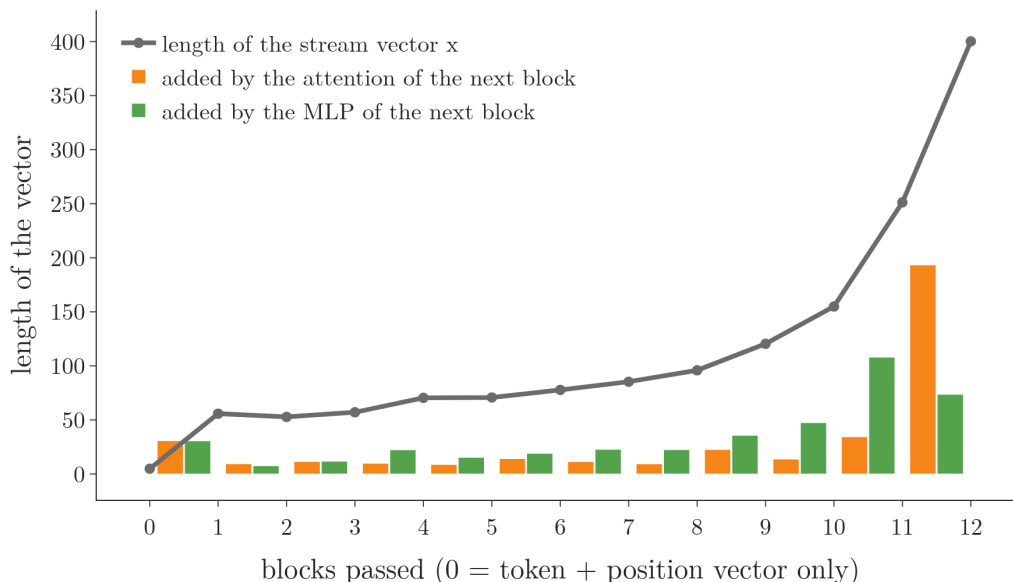


Figure 5: Length of the stream vector of " of" after each block of GPT-2 small (grey), and the lengths of the vectors added by each block's attention (orange) and MLP (green)

- The token-plus-position vector has length 4.9. Block 1 adds two vectors of length about 31 each, and the stream jumps to 56.
- From block 2 to block 10 each addition is short compared with the stream (lengths 8 to 48 against 53 to 155). The direction changes little: the cosine between the stream before and after a block is 0.94 to 0.97.
- Blocks 11 and 12 add the largest vectors (108 from the MLP of block 11, 193 from the attention of block 12). After block 12 the length is 400.

The last LayerNorm matters. With it, the top five next tokens after "Steve Jobs was the founder of" are " Apple", " Microsoft", " the", " IBM", " Intel". Unembedding the raw stream instead, without `ln_f`, gives " the", " a", " ", " and", " ' ": the answer disappears (Notebook).

Extra: Sanderson (2024, Ch 6) pictures attention as computing a change Δe that is **added** to a token's vector, nudging it towards a more specific meaning. In a prenorm model this picture is exact: the stream after a sub-layer is the stream before plus the sub-layer's output, as the formula of section 6.1 says.

6.3 Masked self-attention and the MLP

Key point: The attention sub-layer is the masked multi-head attention of the earlier Notes: 12 heads of 64 in GPT-2 small. The MLP maps each token's 768 numbers to 3,072 and back, the same for every token.

Both parts were built in earlier Notes, so here is only what GPT-2 small uses:

- **Masked multi-head self-attention:** 12 heads, each with queries, keys and values of 64 numbers, $12 \times 64 = 768$ (the [multi-head attention Note](#)). The mask stops each

token from reading later tokens (the [masked self-attention Note](#)). In the Notebook, every attention weight above the diagonal, a future key, is exactly 0.

- **MLP:** $768 \rightarrow 3,072 \rightarrow 768$, applied to each token alone (the [transformer encoder Note](#), §5.3). GPT-1 already used “3072 dimensional inner states” (Radford et al. 2018, §4.1): four times the width. The [MLP Note](#) reads what the rows and columns of these two matrices do.

6.4 GELU

Key point: GPT replaces the ReLU of the MLP with **GELU** (G-834), $x \Phi(x)$, a smooth curve that lets small negative values through.

Radford et al. (2018, §4.1): “For the activation function, we used the Gaussian Error Linear Unit (GELU)”. Hendrycks and Gimpel (2016, §2) define it as the input times the probability that a standard normal value is smaller than it:

$$\text{GELU}(x) = x \Phi(x)$$

where Φ is the standard normal cumulative distribution function. They also give a faster approximation, $0.5x(1 + \tanh[\sqrt{2/\pi}(x + 0.044715x^3)])$, which is what GPT-2 uses (its configuration names it `gelu_new`). On $[-6, 6]$ the two differ by at most 0.0005 (Notebook).

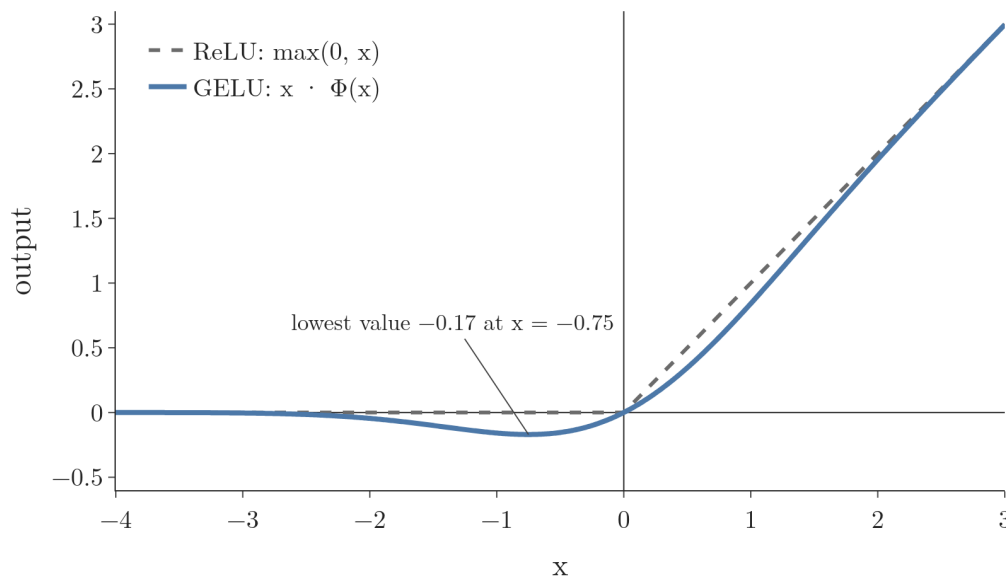


Figure 6: GELU and ReLU. For large positive x both give x ; for large negative x both give about 0; near 0 GELU is a smooth curve that dips to -0.17

Figure 6 shows the difference. ReLU cuts every negative input to exactly 0 with a sharp corner. GELU passes small negative values as small negative outputs, down to -0.17 at

$x = -0.75$, and its curve has no corner. In GPT-2, negative inputs are the common case: on the example prompt, 83.4 percent of the MLP’s 3,072 hidden values are negative before GELU, and their outputs all lie between -0.17 and 0 (Notebook). In the words of Hendrycks and Gimpel (2016, abstract), the GELU “weights inputs by their value, rather than gates inputs by their sign as in ReLUs”.

7. A guess at every position

Key point: After the last block, every position’s vector goes through the final LayerNorm and is multiplied by W_E^T : one score per vocabulary token. One pass therefore gives a next-token guess at every position. Training uses all the guesses; generation uses only the last.

7.1 The tied unembedding

Key point: GPT-2 has no separate output matrix. The scores for the next token are the dot products of the final vector with every row of the token embedding matrix W_E .

GPT-1 wrote the output layer as $P(u) = \text{softmax}(h_n W_e^T)$, with W_e the token embedding matrix (Radford et al. 2018, §3.1, eq. 2). The released GPT-2 file follows the same design: it contains `wte` and no separate output matrix (Notebook). Using one matrix for both jobs is called **tying** the weights (**tied weights**, G-1973), and the translation transformer did the same (Vaswani et al. 2017, §3.4). The [unembedding Note](#) shows what these dot products mean and how a token is chosen from them.

7.2 Every position predicts

Key point: On “Steve Jobs was the founder of”, the same pass guesses the token after every word. After “ founder” it guesses “ of” (0.74, correct); after “ of” it guesses “ Apple” (0.69).

After	Top guess	Probability	True next token	Its probability
Steve	,	0.05	Jobs	below 0.01
Jobs	,	0.15	was	0.05
was	a	0.11	the	0.06
the	first	0.20	founder	0.01
founder	of	0.74	of	0.74
of	Apple	0.69	(not yet written)	

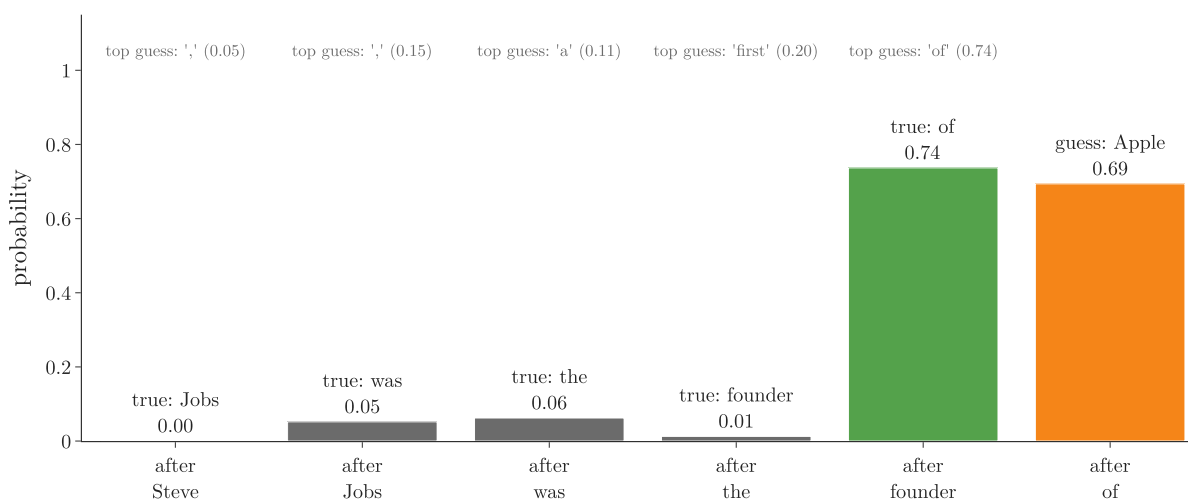


Figure 7: The table as bars: the probability the pass gave the true next token at each position (green when it was also the top guess), with the top guess above. At the last position there is no true token yet; the orange bar is the guess used for generation

Figure 7 draws the table. The early guesses are reasonable but wrong: after “Steve” alone, many continuations are possible. Each position can only see the tokens before it, which is exactly what the mask enforces. The Notebook checks it: replacing the last token ” of” by ” in” changes the last position’s scores by up to 42, and the scores at positions 0 to 4 by exactly 0.

7.3 Training: all positions at once

Key point: The training loss is the average, over all positions, of minus the log of the probability given to the true next token. On a real paragraph GPT-2 small scores 3.69, against 10.83 for a uniform guess.

This is the same training as the [transformer decoder Note](#), §9: one cross-entropy per position (the [loss functions Note](#)), averaged. The Notebook measures it on the first two sentences of the GPT-2 paper’s own abstract, text published after GPT-2’s training data was collected (WebText holds no links created after December 2017; Radford et al. 2019, §2.1):

Measure	Value
Tokens	61, so 60 predictions, all from one pass
Mean cross-entropy	3.69 (natural log)
Uniform guess over 50,257 tokens	$\ln 50,257 = 10.83$
Perplexity (G-1492), $e^{3.69}$	40.1
Top guess correct	36.7 percent

The easiest prediction was ” as” after “tasks, such” (loss 0.006, probability 0.994). The

hardest were content words the context does not force, such as ” begin” in “language models begin to learn” (loss 9.44).

7.4 Generation: predict, pick, append, repeat

Key point: To write text, GPT keeps only the last position’s guess, picks a token, appends it, and runs the whole pass again on the longer text.

This is the loop of the [transformer inference Note](#), without the encoder. Picking the most likely token each time (**greedy decoding**, G-870) on our prompt gives “Steve Jobs was the founder of Apple, and he ...”: ” Apple” (0.69), then “,” (0.31), then ” and” (0.18), then ” he” (0.16).

As a check of our NumPy model, we ran greedy decoding on the prompt of the Hugging Face guide to text generation, which uses the same GPT-2 small weights (von Platen 2020). Our 30 tokens are word for word the guide’s: “I enjoy walking with my cute dog, but I’m not sure if I’ll ever be able to walk with my dog. I’m not sure if I’ll ever be able to walk”. The repetition is a known weakness of always taking the top token; the [unembedding and sampling Note](#) replaces it by sampling.

Sanderson (2024, Ch 5) frames a chatbot as this same loop run on a text that begins with a **system prompt** (G-1935), a few lines setting the scene of a user talking with a helpful assistant, followed by the user’s question. The model then continues the text as the assistant would. Turning a pre-trained model into a good assistant needs further training, which the [history of LLMs Note](#), §9, describes.

8. Context size

Key point: A GPT model reads at most a fixed number of tokens, its **context size** (G-459): 512 for GPT-1, 1,024 for GPT-2, 2,048 for GPT-3. Anything earlier in a longer text cannot affect the prediction.

GPT-2 “increase[d] the context size from 512 to 1024 tokens” (Radford et al. 2019, §2.3); GPT-3 uses “a context window of $n_{\text{ctx}} = 2048$ tokens” (Brown et al. 2020, §2.1). In GPT-2 the limit is built into the weights: W_P has exactly 1,024 rows, so there is no position vector for a 1,025th token, and our forward pass refuses such an input (Notebook). A longer text must be cut, and the cut-off tokens cannot influence the next guess at all.

The window also sets the cost. Each attention head computes one weight for every pair of positions, n^2 of them (the [introduction to transformers Note](#) measures the cost):

Context n	Weights per head, n^2	All 144 heads of GPT-2 small
128	16,384	2,359,296
1,024	1,048,576	150,994,944
2,048	4,194,304	603,979,776

Doubling the context multiplies the attention weights by 4 (Figure 8).

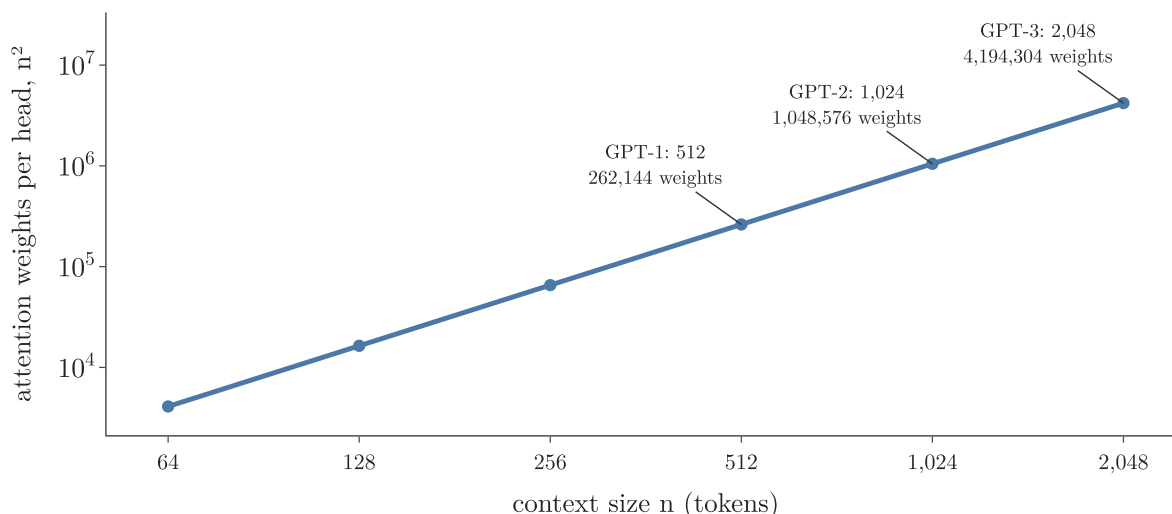


Figure 8: Attention weights computed by one head for a context of n tokens: n^2 , on log scales. Each doubling of the context, as from GPT-1 to GPT-2 to GPT-3, multiplies them by 4

9. Counting the parameters

Key point: GPT-2 small has 124,439,808 parameters: 31 percent in the token embedding, 23 percent in attention, 46 percent in the MLPs. In GPT-3 the embedding is a tiny share; attention holds a third and the MLPs two thirds of its 174.6 billion.

Sanderson (2024, Ch 5–7) counts GPT-3’s weights one kind of matrix at a time, keeping a running tally. Figure 9 does the same for both models: GPT-2 small counted from the released file, and GPT-3 computed from the sizes in Brown et al. (2020, Table 2.1).

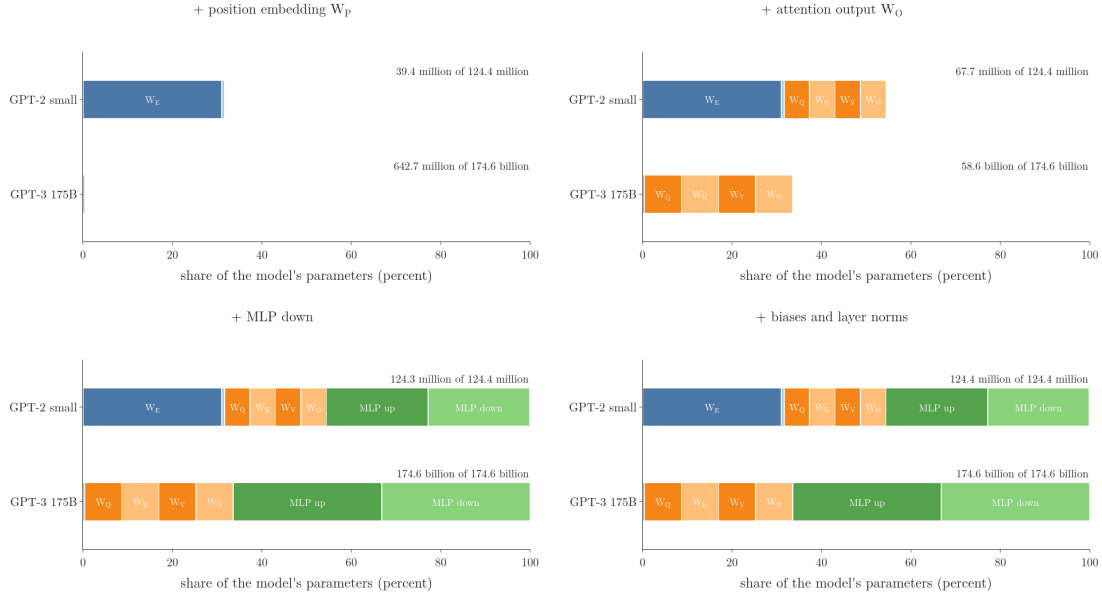


Figure 9: Running tally of parameters, one kind of matrix at a time, as a share of each model. Blue: embeddings. Orange: attention (W_Q, W_K, W_V, W_O). Green: MLP. Grey (thin): biases and LayerNorms

Watch the blue W_E : almost a third of GPT-2 small, a sliver of GPT-3. The vocabulary is the same 50,257 tokens in both, so W_E grows only with the width d , while every block grows with d^2 .

9.1 GPT-2 small, matrix by matrix

Key point: Each block holds $12d^2$ weights plus $13d$ biases and norm values. With $d = 768$, 12 blocks, 50,257 tokens and 1,024 positions this gives exactly the 124,439,808 of the released file.

- In words:** per block, attention has four $d \times d$ matrices (W_Q, W_K, W_V, W_O), and the MLP has a $d \times 4d$ matrix up and a $4d \times d$ matrix down. Biases: $3d$ (Q, K, V) $+d$ (W_O) $+4d$ (MLP up) $+d$ (MLP down); two LayerNorms with a gain and a shift each: $4d$.
- Formula:**

$$N = Vd + n_{\text{ctx}} d + L(4d^2 + 8d^2 + 13d) + 2d$$

The terms, in order: W_E, W_P , then per block attention, MLP and the small vectors, then the final LayerNorm.

- Example:** $V = 50,257, n_{\text{ctx}} = 1,024, d = 768, L = 12$:

Kind of matrix	Shape	Parameters	Share
Token embedding W_E (also the unembedding)	$50,257 \times 768$	38,597,376	31.0 percent

Kind of matrix	Shape	Parameters	Share
Position embedding W_P	$1,024 \times 768$	786,432	0.6 percent
W_Q, W_K, W_V, W_O (12 blocks)	768×768 each	$4 \times 7,077,888$	22.8 percent
MLP up (12 blocks)	$768 \times 3,072$	28,311,552	22.8 percent
MLP down (12 blocks)	$3,072 \times 768$	28,311,552	22.8 percent
Biases and LayerNorms		121,344	0.1 percent
Total		124,439,808	

The formula gives the count of the file to the last parameter (Notebook). The 12 causal-mask buffers stored in the file (`h.N.attn.bias`, $1,024 \times 1,024$ each) are not learned, so they are not counted.

Extra: The GPT-2 paper lists its smallest model as 117M (Radford et al. 2019, Table 2) and says it “is equivalent to the original GPT”. GPT-1 had a BPE vocabulary with 40,000 merges and read 512 tokens (Radford et al. 2018, §4.1). With about 40,000 tokens and 512 positions, the same 12 blocks give about 116 million; with GPT-2’s 50,257 tokens and 1,024 positions they give 124.4 million (Notebook). The 117M label therefore matches GPT-1’s vocabulary and context, not the released GPT-2 file.

9.2 GPT-3

Key point: With $d = 12,288$, 96 blocks and a 2,048-token context, the formula gives 174.6 billion. Attention holds 33 percent, the MLPs 66 percent, the embeddings 0.4 percent.

Brown et al. (2020, Table 2.1) give GPT-3: 96 layers, $d_{\text{model}} = 12,288$, 96 heads of 128, and “the feedforward layer four times the size of the bottleneck layer” (§2.1). They use “the same model and architecture as GPT-2” (§2.1), apart from “alternating dense and locally banded sparse attention patterns”. The formula of section 9.1 gives the count below; it matches the paper’s own counts (next table), so the attention patterns add no weights to the tally:

Kind	Parameters	Share
Token embedding W_E ($50,257 \times 12,288$)	617,558,016	0.35 percent
Position embedding W_P ($2,048 \times 12,288$)	25,165,824	0.01 percent
Attention, 96 blocks	57,982,058,496	33.2 percent
MLP, 96 blocks	115,964,116,992	66.4 percent
Biases and LayerNorms	15,360,000	0.01 percent
Total (tied)	174,604,259,328	

Kind	Parameters	Share
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Is GPT-3’s output matrix tied to W_E , as in GPT-2? The paper does not say it in words, but its own counts do. Its compute appendix lists GPT-3 175B at 174,600 million parameters (Brown et al. 2020, Table D.1). Our tied count is 174,604 million; a separate output matrix would add 617.6 million and give 175,222 million. The same holds for all eight model sizes:

GPT-3 model	Table D.1 (millions)	Our count, tied	Our count, untied
Small (12 layers, $d = 768$)	125	125.2	163.8
Medium (24, 1,024)	356	355.9	407.3
Large (24, 1,536)	760	760.3	837.5
XL (24, 2,048)	1,320	1,315.7	1,418.6
2.7B (32, 2,560)	2,650	2,651.6	2,780.2
6.7B (32, 4,096)	6,660	6,658.4	6,864.3
13B (40, 5,120)	12,850	12,853.4	13,110.7
175B (96, 12,288)	174,600	174,604.3	175,221.8

For the 13B model, Table 2.1 prints $d_{\text{model}} = 5,140$, but its 40 heads of 128 give $40 \times 128 = 5,120$, and only 5,120 reproduces the 12,850 million of Table D.1 (Notebook). The “175B” of the paper’s title is the 174.6 billion rounded.

Extra: Sanderson (2024, Ch 5) adds a separate unembedding matrix of 617 million to the GPT-3 tally. GPT-2’s released weights have no such matrix, and the GPT-3 paper’s own parameter counts match the tied count for every model size, as the table above shows. The difference, 617.6 million, is 0.35 percent of the total, so both versions round to 175 billion.

10. Summary

	Transformer decoder	GPT-2 small
Sub-layers per block	3 (masked self-attention, cross-attention, feed-forward)	2 (masked self-attention, MLP)
LayerNorm	after each addition	at each sub-layer's input, plus one after the last block
Activation	ReLU	GELU
Positions	sine-cosine	learned table, 1,024 rows
Output matrix	tied to the embeddings	tied: the token embedding matrix itself
Size	6 blocks, $d = 512$	12 blocks, $d = 768$, 124,439,808 parameters

- GPT is a transformer decoder without the encoder and without cross-attention.
- Each block adds two changes to the residual stream: attention's, then the MLP's. Prenorm: LayerNorm comes before each sub-layer.
- One pass gives a guess at every position. Training averages the cross-entropy of all positions; generation uses the last guess, appends it and repeats.
- The context size (1,024 for GPT-2) is the most text the model can read; the attention cost grows with its square.
- GPT-2 small: 124,439,808 parameters, a third of them in the token embedding. GPT-3: 174.6 billion, a third in attention and two thirds in the MLPs.

Extra: Three ideas of this Note come from Sanderson's *Transformers, the tech behind LLMs* (3Blue1Brown, 2024, Ch 5): counting the parameters matrix by matrix as a running tally, the colour rule (learned weights in blue or red, data in grey), and generation as "predict, sample, append, repeat". The figures and animations are our own design, made with the released GPT-2 small weights.

11. Sources

Built from

- Sanderson, G. (3Blue1Brown), "Transformers, the tech behind LLMs | Deep Learning Chapter 5", 2024, 3blue1brown.com/lessons/gpt, <https://www.youtube.com/watch?v=wjZofJX0v4M>
- Sanderson, G. (3Blue1Brown), "Attention in transformers, step-by-step | Deep Learning Chapter 6", 2024, 3blue1brown.com/lessons/attention, <https://www.youtube.com/watch?v=eMlx5fFN>
- Radford, A., Narasimhan, K., Salimans, T. and Sutskever, I. (2018). Improving Language Understanding by Generative Pre-Training. OpenAI. §3.1 (eq. 1, the language-model objective; eq. 2, $h_0 = UW_e + W_p$ and $P(u) = \text{softmax}(h_n W_e^T)$); §4.1 "Model specifications" (12-layer decoder-only transformer, 768 states, 12 heads, 3,072 inner states, 512-token sequences, 40,000 BPE merges, GELU, learned position embeddings).

- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D. and Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners. OpenAI. §2.1 (WebText, no links after December 2017); §2.2 (byte pair encoding); §2.3 (LayerNorm moved to the input of each sub-block, extra final LayerNorm, vocabulary 50,257, context 1,024); Table 2 (117M for the smallest model).

Other references

- Brown, T. B. et al. (2020). Language Models are Few-Shot Learners. *NeurIPS 2020*. arXiv:2005.14165 (v4). §2.1 (same architecture as GPT-2, alternating sparse attention, $d_{\text{ff}} = 4d_{\text{model}}$, $n_{\text{ctx}} = 2048$); Table 2.1 (sizes of the 8 models); Appendix D, Table D.1 (parameter counts in millions).
- Vaswani, A. et al. (2017). Attention Is All You Need. *NeurIPS 2017*. arXiv:1706.03762. §3.1 (decoder block); §3.4 (shared weights between the embeddings and the pre-softmax layer).
- Hendrycks, D. and Gimpel, K. (2016). Gaussian Error Linear Units (GELUs). arXiv:1606.08415. Abstract and §2 ($x\Phi(x)$, the tanh approximation).
- von Platen, P. (2020). How to generate text: using different decoding methods for language generation with Transformers. Hugging Face blog, 1 March 2020, huggingface.co/blog/how-to-generate. Greedy search section (GPT-2 output for “I enjoy walking with my cute dog”).
- Weights and tokenizer: `openai-community/gpt2` (`model.safetensors`, `tokenizer.json`), [huggingface.co](https://huggingface.co/openai-community/gpt2).

12. Key terms

Term	Meaning
GPT	Generative Pre-trained Transformer: a decoder-only transformer trained to predict the next token
Decoder-only transformer	A transformer with no encoder and no cross-attention; each block has masked self-attention and an MLP
Token	A piece of text from a fixed vocabulary: a word, part of a word or a symbol; GPT-2 has 50,257
Byte pair encoding (BPE)	A way to build a vocabulary by repeatedly merging the most frequent pair of symbols
Token embedding matrix W_E	The learned table with one vector per token; in GPT-2 also used as the output matrix
Position embedding matrix W_P	The learned table with one vector per position; 1,024 rows in GPT-2
Residual stream	The token vector that passes from block to block, to which every sub-layer adds its output

Term	Meaning
Prenorm	Placing LayerNorm at the input of each sub-layer instead of after the addition
GELU	Gaussian Error Linear Unit, $x \Phi(x)$: a smooth activation used in GPT's MLPs
Tied weights	Using the same matrix for the token embedding and the output layer
Context size	The largest number of tokens the model can read at once: 1,024 for GPT-2, 2,048 for GPT-3
Greedy decoding	Always picking the most likely next token
Perplexity	The exponential of the mean cross-entropy per token; lower is better
System prompt	Text placed before the user's message that sets the scene for a chatbot